

Insight Fusion – Creating comprehensive data sets

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Abstract

This assignment clearly states an analysis of a student performance dataset, which is obtained from Kaggle, and this explores key insights through data visualization techniques. The study demonstrates the different concepts used for the data analysis and also how pivot tables, charts, and graphs can be used to summarize patterns, highlight trends, and support data-driven interpretation. By altering this raw student dataset from Kaggle into the clear and meaningful visual representations, the assignment describes how these analytical tools can enhance the understanding of student outcomes and provide a foundation for the informed decision-making.

Correlation

Correlation chosen for study time and G3 fields

studytime ↔ G3 (final grade)

Pearson $r = 0.098$ (very weak positive linear relationship), $n = 395$.

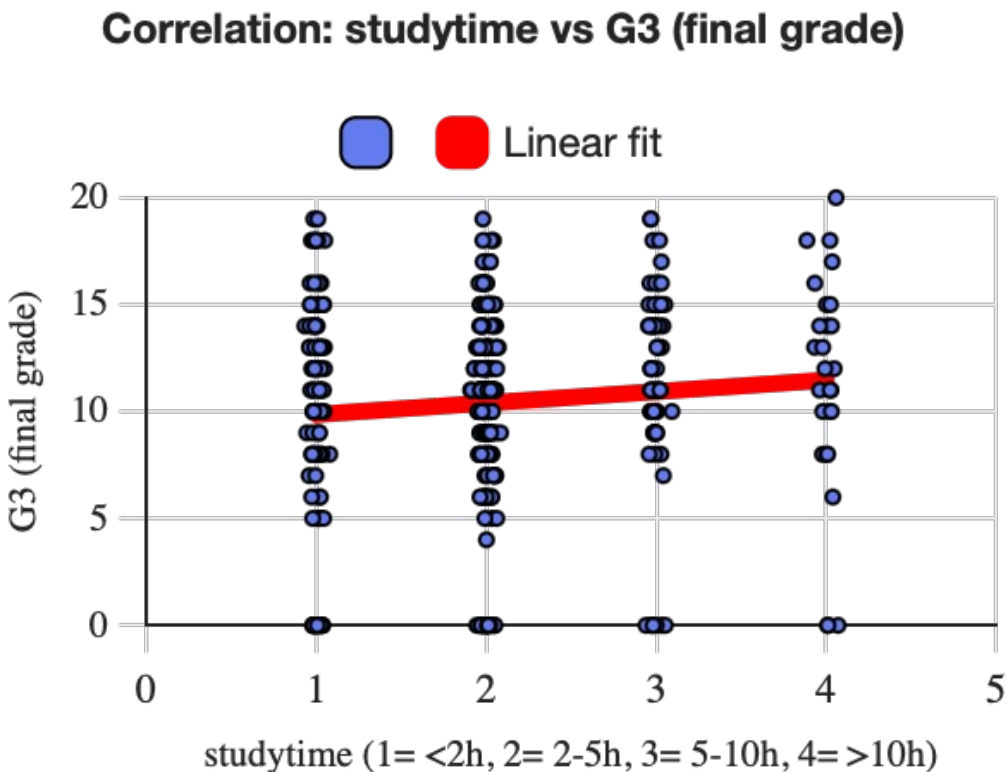
Spearman $\rho = 0.105$ (very weak positive monotonic relationship).

Interpretation: In this dataset, students who report more weekly study time tend to have slightly higher final grades, but the relationship is very small much of the variation in G3 isn't explained by studytime alone. (Correlation near 0 $\Rightarrow$ negligible linear association.)

Visual representation:

Figure 1

Correlation of Study Time and G3



Pearson Correlation (r)

Formula

Where:

- = studytime values
- = G3 (final grade)
- students with valid values
- = mean of studytime
- = mean of G3

The Python code used for Pearson correlation was:

Python

```
pearson_r = sub['studytime'].corr(sub['G3'], method='pearson')
```

Interpretation

- Direction: Positive → studytime ↑ → grades slightly ↑
- Strength: Very weak (close to 0)
- Meaning: Study time explains almost no variation in G3 by itself.

Are there any apparent causal relationships

Interpretation

These patterns are almost consistent with the idea that prior performance is the dominant predictor of performance results in this data, and that other factors (studytime, absences, alcohol

variables, social time) show minimal incremental association with the final grade once we account for prior grade but none of this proves causation.

Confounding

Many variables that affect both the “treatment” and the outcome may be unobserved (e.g., motivation, baseline ability, teacher effects, family expectations).

- Reverse causality: Students who are already doing well might (a) study differently, (b) miss classes for sanctioned activities, or (c) receive different teacher attention so the direction of influence can be ambiguous.
- Measurement & selection: Self-reported study time is coarse (1–4 buckets). Absence recording may vary by context. These issues can attenuate or distort estimated effects.
- Cross sectional snapshot: We do not observe policy changes or exogenous shocks that would help isolate cause-and-effect from mere correlation.

Holistic Understanding of The Datasets

The core structure of performance: strong persistence across terms

The prior period grades are by far the strongest signals for the final grade (G3):

- $G2 \leftrightarrow G3: r \approx 0.905$, $G1 \leftrightarrow G3: r \approx 0.801$ (very strong, positive). This means performance is highly persistent; in a simple bivariate sense, G2 alone would explain roughly of variance in G3.

Why this matters holistically: Most of what we will observe in the final grade is already “in motion” by the penultimate terms. Any assessment, support, or intervention strategy should leverage earlier grades first; they carry most of the predictive signal in this dataset.

“Common suspects” are present but weak on their own.

- Studytime ↔ G3: $r \approx 0.098$ (very small positive).
- Absences ↔ G3: $r \approx 0.034$ (essentially zero). Even after trimming the top 5% of very high absences, $r \approx 0.083$ —still tiny.

Why this matters holistically: Simple narratives like “more study time = much higher grades” or “more absences = much lower grades”. Any policy or support action based solely on these two variables would have limited impact without considering context like prior achievement.

Pivot Table and Charts

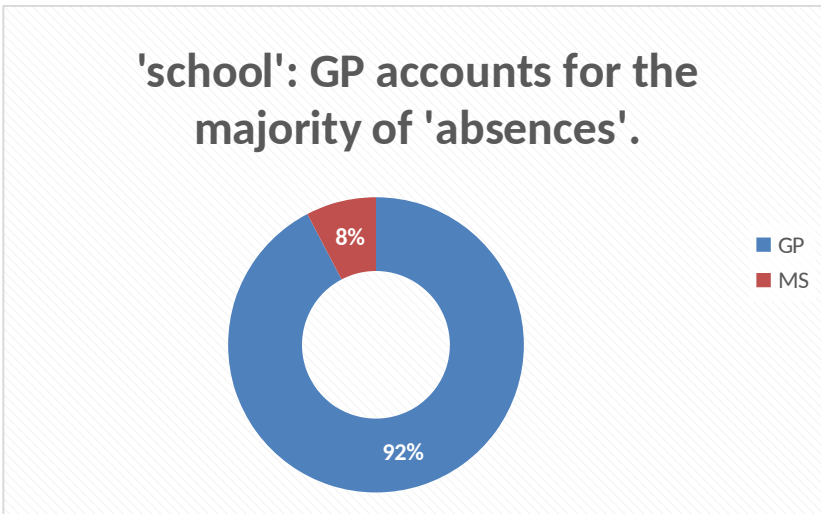
1. Generated A Pie Chart for The Type of School and Absences

Pivot table

Sum of absences	
school	
GP	2082
MS	173
Grand Total	2255

Figure 2

Pie Chart of GP Accounts for The Majority of Absences

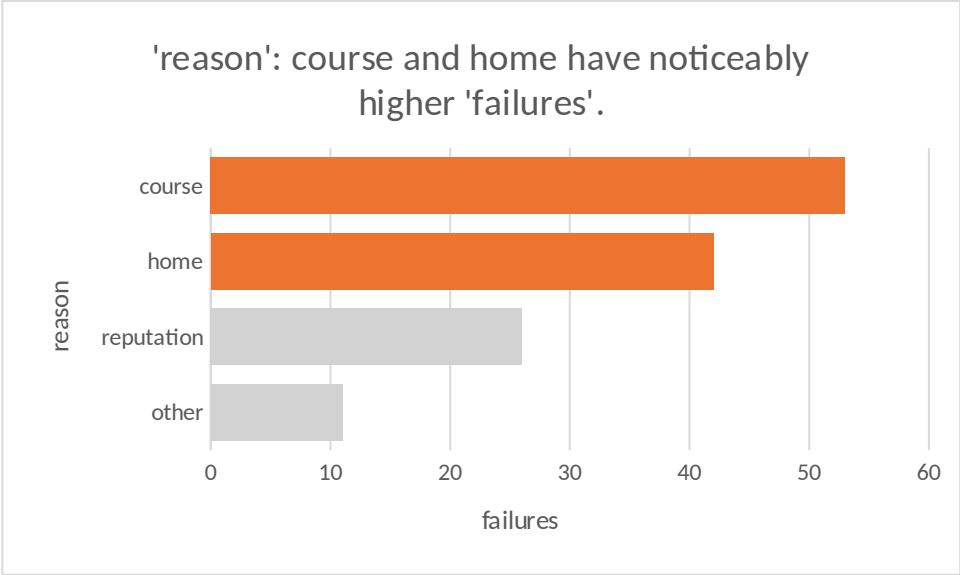


2. Generated a bar chart for the failures with different reasons.

reason	Sum of failures
course	53
home	42
reputati	
on	26
other	11
Grand Total	132

Figure 3

Bar Graph of Different Reasons for The Failures



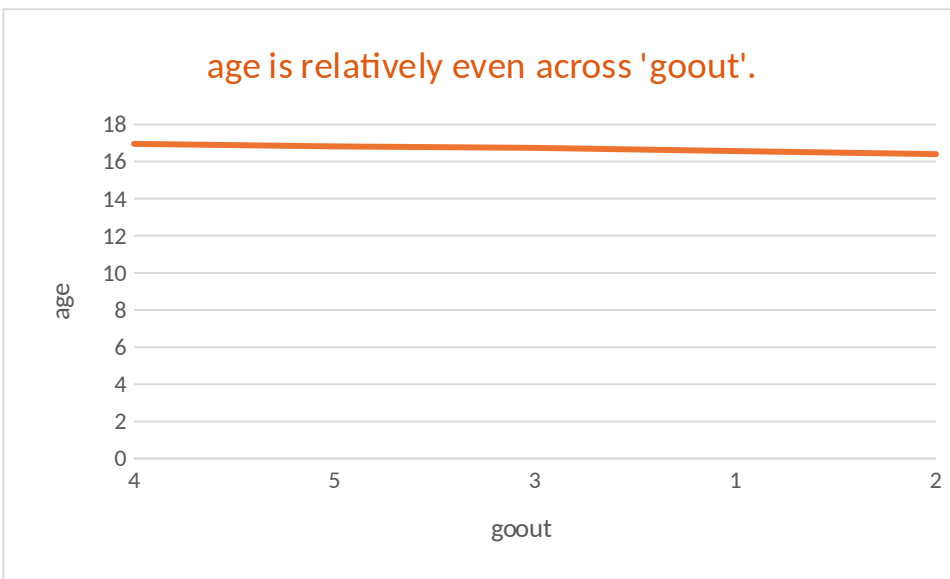
3. Generated A Line Graph Which Demonstrates Students Who Go Out Based on Their Age

Average of age	
goout	
	16.953488
4	37
	16.811320
5	75
	16.738461
3	54
	16.565217
1	39
	16.398058
2	25
Grand	16.69620

Total	253
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Figure 4

Line Graph of Student Age and Students Who Goes Out



Conclusion

This analysis of the Kaggle student performance dataset highlights how data visualization techniques such as pivot tables, charts, and graphs which can transform raw data into meaningful insights. By determining the relationships between different variables like absences, study time, prior grades, social factors, and final performance, the assignment demonstrates the importance of selecting appropriate visuals to uncover trends and support interpretation. Overall, this study reinforces the value of thoughtful visualization practices in enhancing clarity, supporting decision-making, and deepening understanding of complex datasets.

References

Copilot is used for final draft of this essay

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